

SOLA STAVANGER, Norway

WMO: 014150

Lat: 58.884N

Lon: 5.637E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.8	-6.5	-14.1	1.1	-5.3	-11.9	1.4	-4.3	14.6	3.5	13.4	3.7	1.4	140	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	24.3	17.3	22.2	16.5	20.4	15.9	18.6	22.5	17.5	20.9	16.6	19.5	5.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.2	20.1	16.1	11.4	19.1	15.2	10.8	18.2	52.5	22.5	49.3	20.8	46.5	19.4	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	11.4	10.2	-10.7	27.6	2.6	2.1	-12.6	29.1	-14.1	30.3	-15.6	31.5	-17.5	33.0
			-11.1	19.9	2.7	1.5	-13.1	21.0	-14.7	21.9	-16.2	22.8	-18.1	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	2.1	1.9	3.5	6.9	10.0	12.8	15.5	15.7	13.2	9.2	5.5	3.0
	DBStd	5.84	3.68	3.43	2.85	2.91	3.07	2.50	2.65	2.55	2.46	2.82	3.44	4.13
	HDD10.0	1216	244	226	202	102	35	2	0	0	3	47	138	217
	HDD18.3	3677	503	459	460	344	258	167	98	89	154	283	386	476
	CDD10.0	603	0	0	0	8	37	87	169	178	100	22	2	0
	CDD18.3	21	0	0	0	0	1	2	8	9	1	0	0	0
	CDH23.3	103	0	0	0	0	5	11	48	35	3	0	0	0
	CDH26.7	11	0	0	0	0	0	1	7	3	0	0	0	0
Wind	WSAvg	4.5	5.2	4.7	4.7	4.4	4.5	4.4	4.0	4.1	4.3	4.8	4.6	4.7
Precipitation	PrecAvg	1199	103	75	72	56	65	69	90	122	143	144	139	120
	PrecMax	1733	236	177	200	126	168	161	244	241	328	299	244	248
	PrecMin	841	4	1	2	3	2	16	21	5	27	37	25	8
	PrecStd	193	60	43	46	26	35	33	52	52	56	61	56	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	8.9	12.5	19.3	23.5	25.0	27.7	26.7	23.2	17.3	13.3	10.6
		MCWB	8.2	7.6	7.9	11.4	15.1	15.5	18.4	18.5	16.7	13.6	11.0	9.3
	2%	DB	8.2	7.9	9.4	15.8	20.2	21.4	24.4	24.0	20.1	15.2	11.9	9.7
		MCWB	7.2	7.1	6.7	9.6	13.4	15.1	17.8	17.8	15.2	12.6	10.1	8.6
	5%	DB	7.5	7.2	8.0	13.5	17.5	18.9	21.9	21.8	18.2	14.1	11.0	8.9
		MCWB	6.5	6.4	6.2	8.8	12.3	14.3	16.9	17.1	14.5	11.9	9.5	7.8
	10%	DB	6.9	6.6	7.2	11.6	15.1	17.0	20.0	20.1	17.0	13.1	10.1	8.1
		MCWB	5.9	5.8	5.7	8.0	11.1	13.6	16.4	16.4	14.0	11.1	8.7	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	8.0	8.9	12.7	16.5	17.4	19.9	20.1	17.5	14.4	11.8	9.7
		MCDB	8.8	8.3	10.5	17.2	21.7	21.0	24.8	23.1	20.7	16.0	12.7	10.1
	2%	WB	7.5	7.3	7.4	10.7	14.2	16.0	18.7	18.7	16.2	13.2	10.6	8.8
		MCDB	7.9	7.7	8.7	14.3	18.6	19.8	22.9	22.2	18.8	14.7	11.6	9.4
	5%	WB	6.8	6.6	6.7	9.6	12.7	14.9	17.6	17.7	15.3	12.5	9.8	8.0
		MCDB	7.3	7.1	7.6	12.6	16.5	18.3	21.1	21.0	17.7	13.8	10.6	8.7
	10%	WB	6.0	5.9	6.0	8.5	11.4	13.9	16.6	16.7	14.4	11.6	8.9	7.2
		MCDB	6.6	6.5	6.9	11.0	14.6	16.6	19.5	19.5	16.6	12.9	9.9	8.0
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.7	5.3	6.2	6.3	5.7	5.7	5.8	5.5	5.2	4.7	4.5
		MCDBR	4.1	3.8	6.0	9.6	10.5	9.4	9.7	8.6	7.3	5.8	4.5	3.8
		MCWBR	4.0	3.6	4.3	5.4	5.6	5.1	4.9	4.5	4.3	4.2	4.2	4.0
	5% WB	MCDBR	4.0	3.6	5.1	8.0	9.4	8.2	8.4	7.5	6.3	4.7	4.4	3.8
		MCWBR	4.1	3.6	4.2	4.9	5.6	4.9	4.7	4.4	4.4	4.1	4.5	4.1
Clear-Sky Solar Irradiance	taub	0.309	0.332	0.352	0.383	0.372	0.365	0.379	0.380	0.361	0.341	0.313	0.287	
	taud	2.177	2.278	2.320	2.291	2.405	2.454	2.434	2.430	2.468	2.410	2.256	2.126	
	Ebn at Noon	485	679	783	819	860	871	847	818	774	669	483	373	
	Edh at Noon	48	71	91	110	105	102	102	95	79	63	44	35	
All-Sky Solar Radiation	RadAvg	0.30	0.90	2.12	3.82	5.17	5.56	4.94	3.81	2.45	1.17	0.41	0.19	
	RadStd	0.05	0.15	0.24	0.37	0.57	0.62	0.49	0.32	0.21	0.14	0.06	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-137	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A

Nomenclature: See separate page